

ITA: 0501 COMPUTER VISION

CO:1-AT:1-Application  
based Question

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1. Problem: Small objects appear unclear because the image has insufficient sampling rate.

Application / Solution: using sampling and a camera with high resolution so the small objects appear clearly.

2. Problem: At night the image captured using a surveillance camera is dark and unclear.

Application / Solution: using histogram equalization so that the dark image ~~is~~ increased the contrast and brightness. And also use filtering to remove unnecessary noise.

3. Problem: A medical image contains continuous intensity values and to be converted to digital values.

Application / Solution: using Quantization, it converts brightness values into digital values (0-255) so that the computer can store and process.

4. Problem: Defects are not visible because some parts of the product in image is dark.

Application / Solution: Contrast enhancement is used as it improves the contrast of the dark areas and the cracks become more visible.

5. Problem:

To separate the objects captured by a self-driving car.

Application / Solution:

using image segmentation so that it divides an image into different parts so that it can detect obstacles and pedestrians to drive safely.

6. Problem:

To identify the cracks on manufactured components

Application / Solution:

Edge detection are used as it makes the cracks or defects more visible to find as the cracks appear as thin lines.

7. Problem:

A medical image having a tumor is not clear because of noise

Application / Solutions:

Median Filtering is used as it removes unwanted noise without disturbing the important edges and boundaries, making the tumor more visible.

Problem:

A traffic monitoring camera captures a bad lighting image.

Application / Solution:

Object detection techniques like YOLO is used so that it identifies the vehicles and also tracks the vehicle.

9. Problem:

A camera captures a blurry image while moving.

Application / Solution:

Image processing techniques like image deblurring is used as it removes the unwanted blur and enhances the quality of the image.

10. Problem:

An industrial vision system must distinguish defective objects from good products.

Application / Solution:

Image classification is used so that it can classify the defective products from the good products for improving speed, accuracy and reduces the manual work.